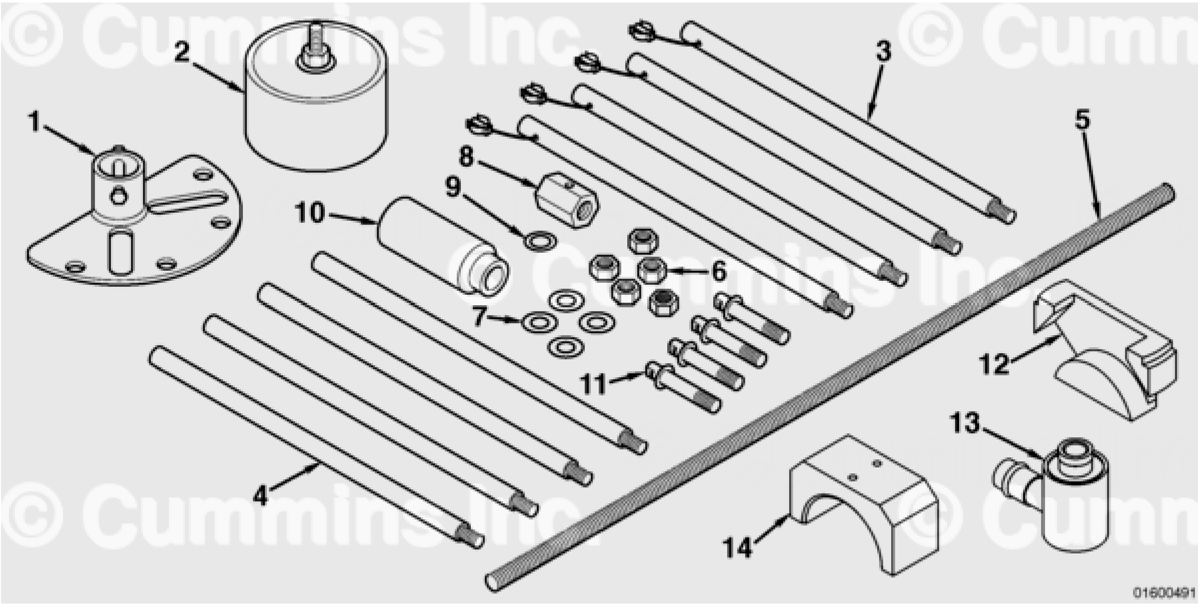
Trainings

General Information



For engines containing the straight split connecting rod, the cylinder liner **must** be removed as an assembly with the piston and connecting rod.

Power Cylinder Install/Removal Tool, Part Number 4919227, is required for this process.

The plunger fastens to the cylinder liner and aids in extracting the piston, liner, and connecting rod (power cylinder) as an assembly.

The tool consists of the following components:

1. Top plate
2. Plunger
3. Frame tube legs
4. Extension frame tube legs
5. Threaded bar
6. Securing and extraction nuts
7. Securing washers
8. Hexagonal coupling
9. Thrust washer
10. Socket
11. Block guide studs
12. Cylinder liner removal tool
13. Hydraulic cylinder
14. Locating cup.

Additional service tools required to assemble and install the power cylinder are:

- Connecting rod guide pins, Part Number 4918922 (X4)
- Piston ring compressor, Part Number 4918898 (X1).

Note : There are two different piston and liner combinations available for this engine platform. The FCD piston **must** be used in combination with a nitrided cylinder liner. The steel piston **must** be used in combination with a high boron liner. It is important that these parts are used in this combination stated and **never** mixed. An engine **must never** be run with a mixture of FCD and steel pistons; It **must** contain **only** a full set of FCD pistons or a full set of steel pistons

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid contact of hot oil with your skin.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)

- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Remove the lubricating oil pan suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Remove the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Remove and discard the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)
- Remove the piston cooling nozzles for the cylinders requiring removal. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-046-tr.html)

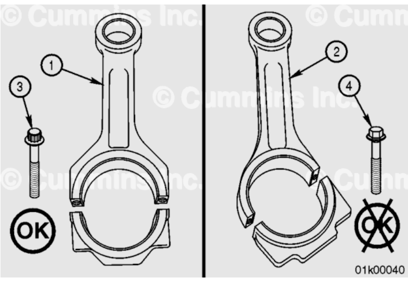
Note : The block stiffener plate has been discontinued for QSK45 and QSK60 engines.

Remove

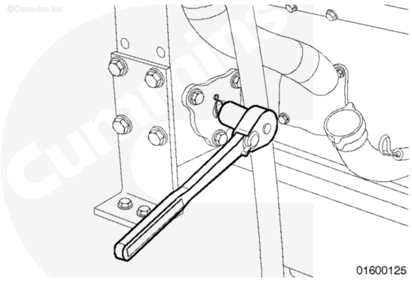
The following steps **must** be used for engines containing the straight split connecting rod **only** (1).

Use the following procedure for angle split connecting rods (2). Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

The straight split connecting rod can be identified by 12 point head capscrews (3).

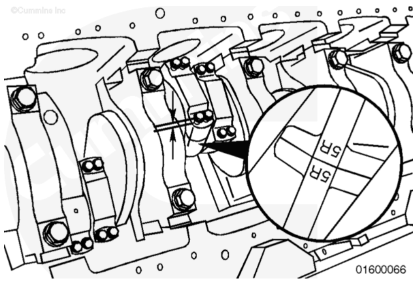


Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.



The connecting rods **must** have the cylinder number and bank marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for the correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

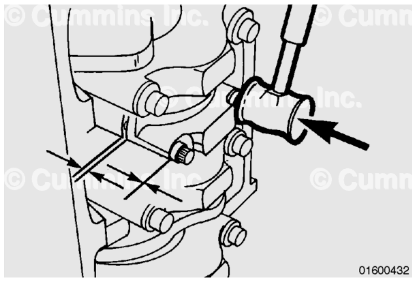
Loosen the capscrews until there is 6 mm [0.236 in] of clearance between the rod cap and the capscrew head.



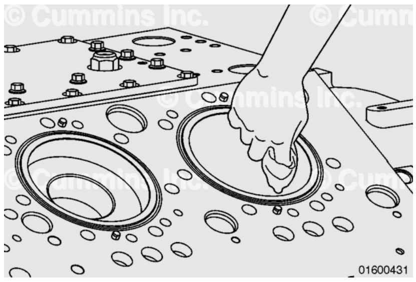
Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate.

Tighten the capscrews by hand.

This step is used to break the connection between the connecting rod and cap. **Do not** remove the cap at this time.

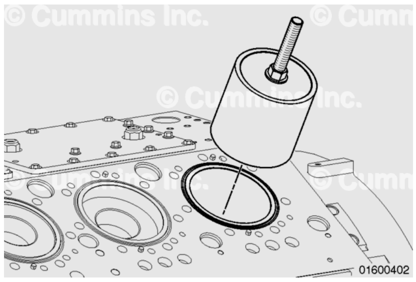


Remove any excess residual oil from the liner bore.



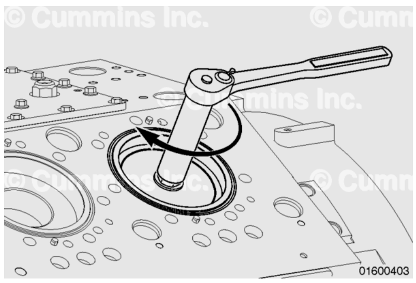
Insert the plunger into the cylinder liner until it contacts the piston.

Once inserted, the top face of the plunger will be approximately 20 mm [0.79 in] below the top of the liner.

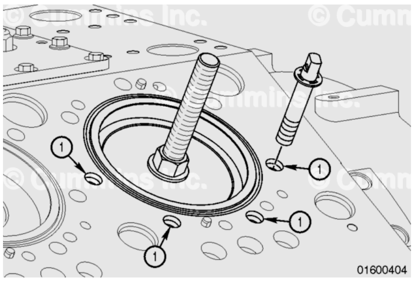


Tighten the plunger to lock the plunger into the liner.

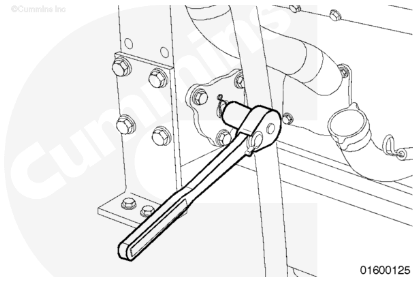
Torque Value: 125 n•m [92 ft-lb]



Insert the four block guide studs into the block at the locations shown (1) until the circlip contacts the block face.

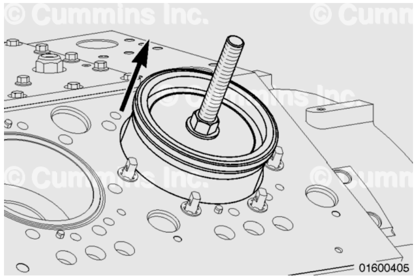


Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at top dead center.



As the connecting rod moves up to top dead center, the power cylinder assembly is pushed out of the block.

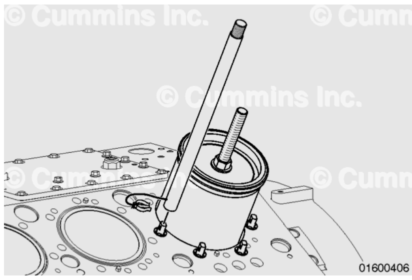
If the interference fit between the liner and the block is excessive, the plunger may be pushed up without removing the liner. If this occurs, reference the "Hydraulic Power Cylinder" section below for power cylinder removal.



The tooling described in the following section of the procedure is for lifting **only**. It should **not** be used to break the cylinder block adhesion to the upper press fit area and packing bore seals.

Install the four frame tube legs and lock them to the guide studs with the attached pins.

Note : In applications with sufficient overhead clearance, the additional extension frame tube legs can also be fitted.

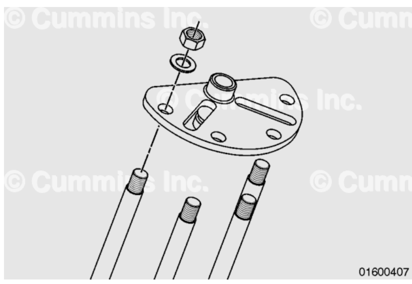


⚠ CAUTION ⚠

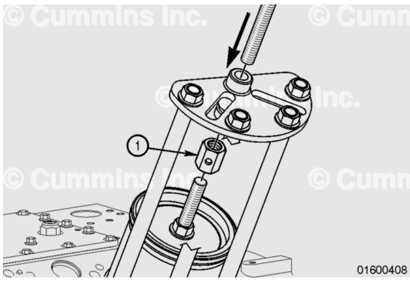
Do not over-tighten the nuts or tool damage can occur.

Attach the top plate with the boss facing upwards.

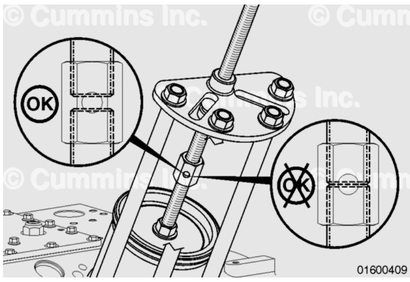
Secure the plate with the four nuts and washers provided.



Insert the threaded bar through the top plate boss and attach it to the plunger using the hexagonal coupling (1).

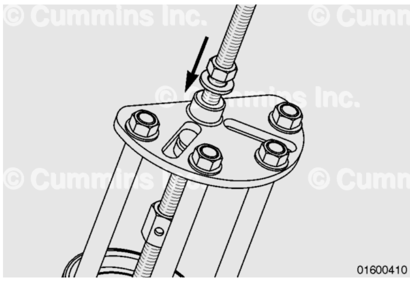


There should be clearance through the locking pin hole in the hexagonal coupling.



Install the thrust washer and extraction nut to the threaded bar. Hand-tighten against the top plate.

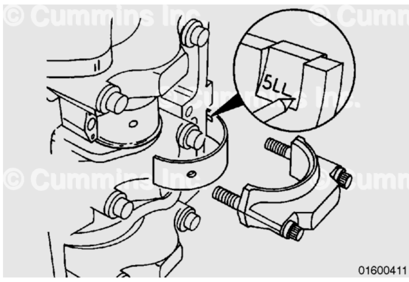
The power cylinder is now locked in place.



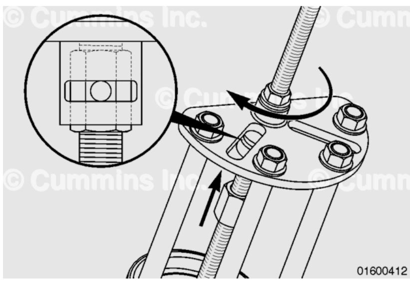
Remove the connecting rod capscrews and the cap.

Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

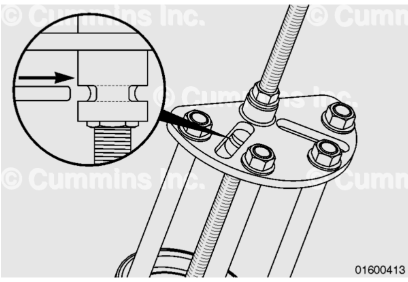
Mark the bearing for future identification or analysis.



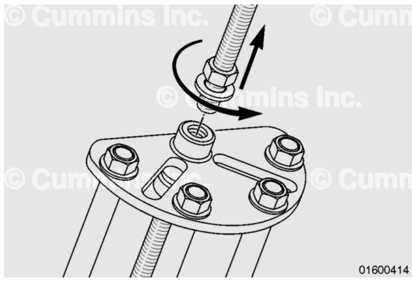
Turn the top nut **clockwise** to draw the power cylinder from the block until the hole in the hexagonal coupling aligns with the hole on the underside of the top plate.



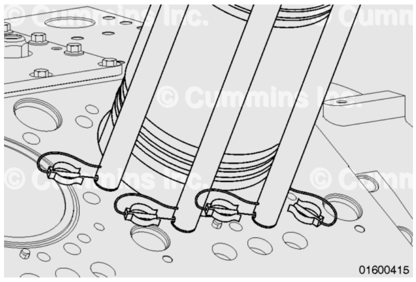
Insert the locking pin through the plate and hexagonal coupling.



Loosen the top nut and remove the threaded bar from the hexagonal coupling and top plate.

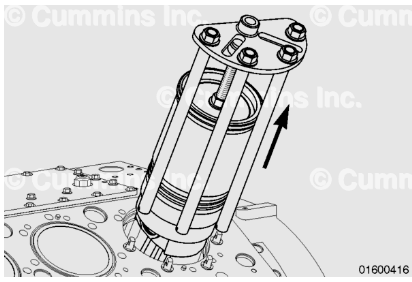


Remove the four locking pins from the lower guide studs.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



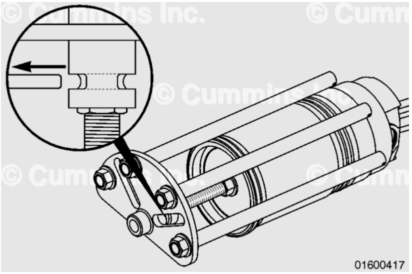
⚠ CAUTION ⚠

Piston ring or liner damage could relieve the vacuum holding the piston in place causing it to fall from the assembly resulting in personal injury or component damage.

Remove the power cylinder assembly.

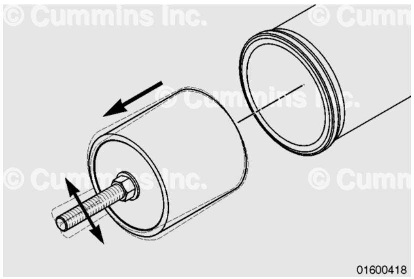
Place the assembly on a pallet or bench.

Remove the top locking pin from the hexagonal coupling to release the lifting assembly.



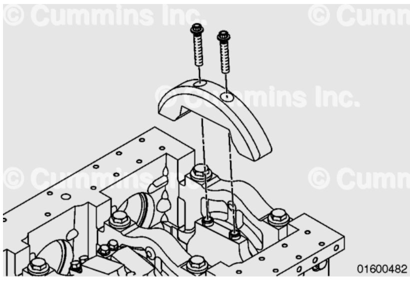
Loosen the plunger nut and gently pull to release it from the liner bore. A side-to-side motion may be required to break the seal between the liner and plunger.

Remove the plunger from the liner.



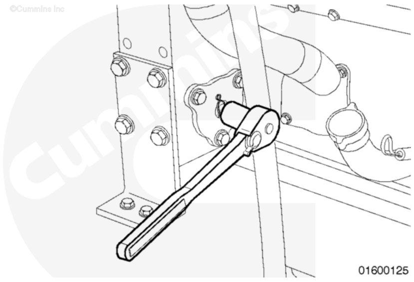
Hydraulic Power Cylinder

Working from the bottom of the engine, remove the counterweight, if required. Refer to Procedure 001-016 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html)

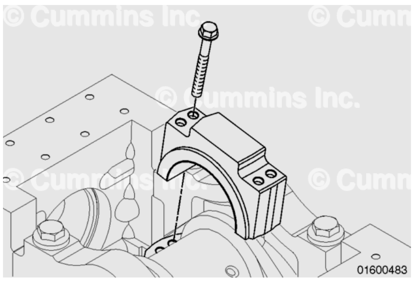


Use the barring mechanism to rotate the engine.

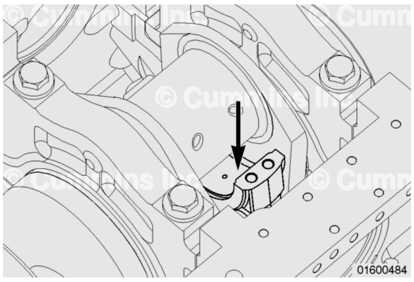
Rotate the crankshaft to position the connecting rod at bottom dead center.



Remove the connecting rod capscrews and the rod cap.

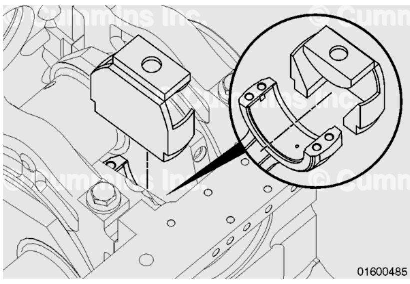


Push the piston and connecting rod away from the crankshaft.

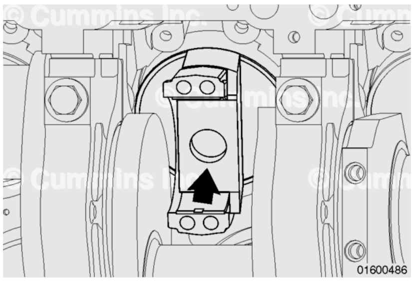


Attach a cylinder liner removal tool to the connecting rod.

Note : The crank may need to be rotated slightly to improve access.



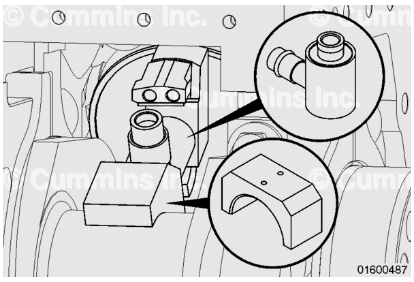
Push the connecting rod and piston all the way up the piston liner until the recess of the cylinder liner removal tool engages the liner.



If rotated earlier, rotate the crankshaft back to bottom cead center.

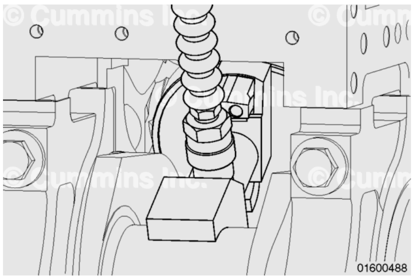
Install the locating cup and hydraulic cylinder onto the crankshaft connecting rod journal.

Note : The locating cup and hydraulic cylinder are bolted together as one piece.

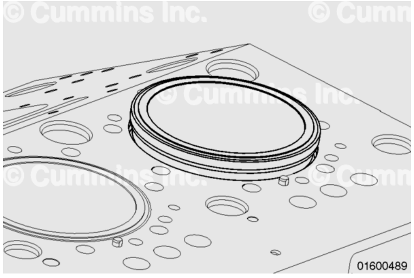


Attach a hydraulic line to the hydraulic cylinder.

Gently apply pressure until the hydraulic ram engages with the cylinder liner removal tool.



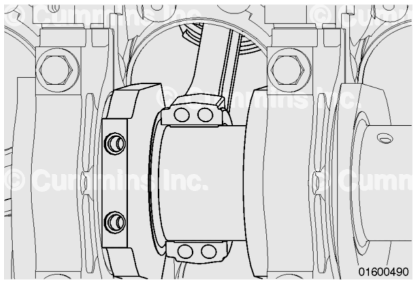
Press the liner out until the liner interference fit is cleared and the liner is loose.



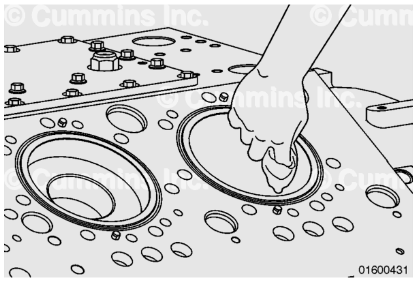
Remove the hydraulic cylinder, remover plate, and cylinder liner removal tool from the engine.

With these removed, push the piston back down the liner and guide the connecting rod back onto the crankshaft. The crankshaft should be at bottom dead center.

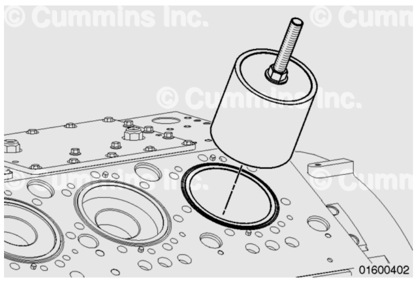
Note : The connecting rod must be pushed down to clear the liner for removal of the cylinder liner removal tool.



Remove any excess residual oil from the liner bore.



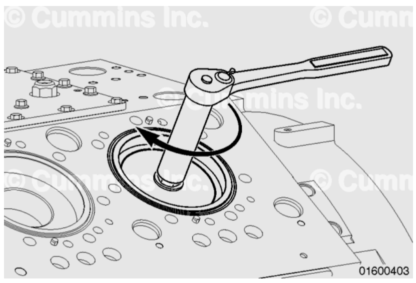
Insert the plunger into the cylinder liner until it contacts the piston.



Tighten the plunger to lock the plunger into the liner.

Torque Value: 125 n•m [92 ft-lb]

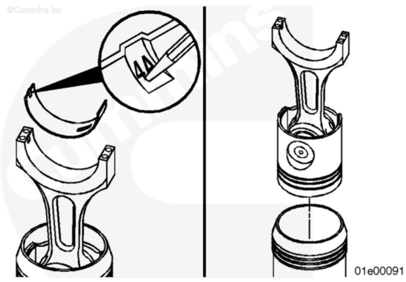
Continue to remove the power cylinder as described above.



Disassemble

Remove and mark the connecting rod bearing with the bank from which the bearing is removed for reassembly (e.g. 4A or 4B).

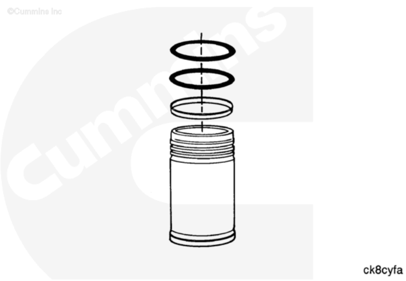
Remove the piston and connecting rod from the bottom of the liner.



Remove the two D-rings.

Remove the crevice seal.

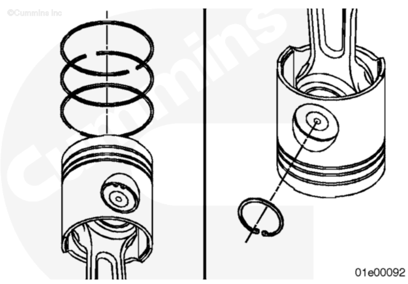
Remove the liner counterbore sealing ring.



Use ring expander pliers, Part Number 3164401, to remove the piston rings.

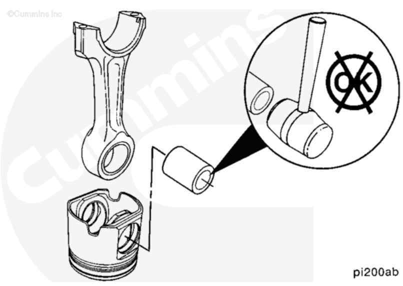
Use retaining ring pliers, Part Number 3164403, to remove the snap ring or circlip.

Mark or tag the rings from each cylinder with the cylinder number and bank for analysis, if desired.



⚠ CAUTION ⚠

Do not drop the piston while removing the piston pin or damage to the piston could occur.



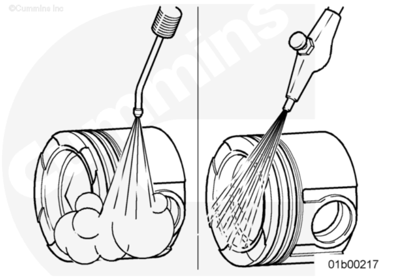
Remove the piston pin from the piston assembly.

The piston and the connecting rod can now be separated.

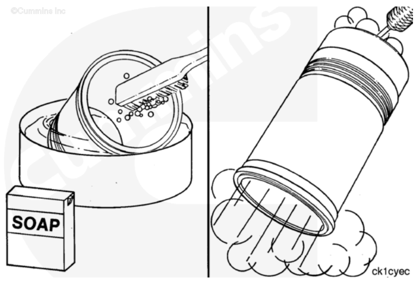
It is recommended that these parts be kept together for possible analysis.

Clean and Inspect for Reuse

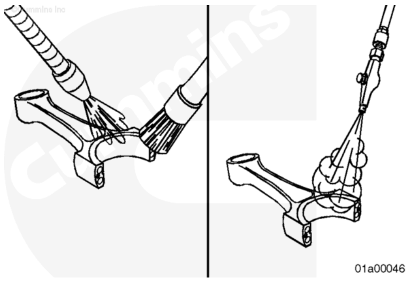
Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-043-tr.html)



Clean and inspect the liners for reuse. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)

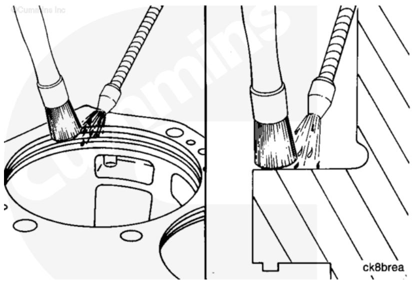


Clean and inspect the connecting rods for reuse. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-014.html)



⚠ **WARNING** ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

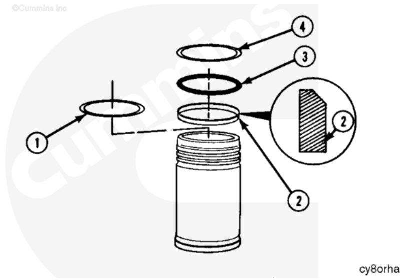


Clean the bottom of the cylinder liner flange with safety solvent.

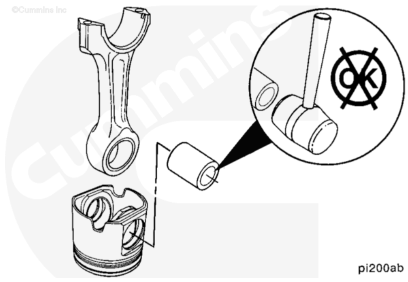
Inspect the cylinder block counterbore and lower packing ring bore. Refer to Procedure 001-027 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-027.html)

Assemble

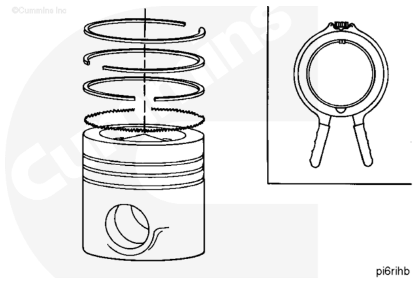
Assemble the D-rings (o-rings) (3)(4), crevice seal (2), and sealing ring (1) to the liner. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)



Assemble the piston to the connecting rod. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

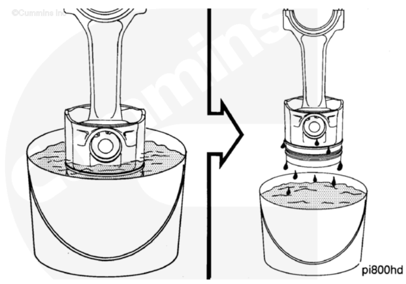


Assemble the piston rings onto the piston. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-047-tr.html)

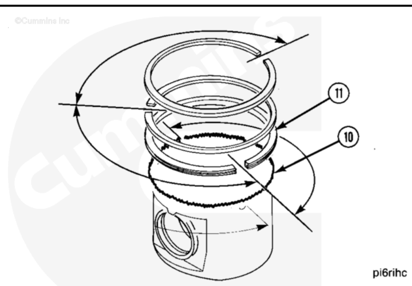


Immerse the piston in engine oil until the rings are covered.

Allow the excess oil to drip off the assembly.

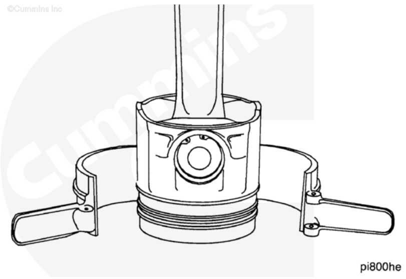


Check to make sure the ring gap position is still correct.



⚠ CAUTION ⚠

Verify that the oil ring is correctly seated during assembly for the steel piston. Failure to do so can lead to damage during assembly with the cylinder liner.



⚠ CAUTION ⚠

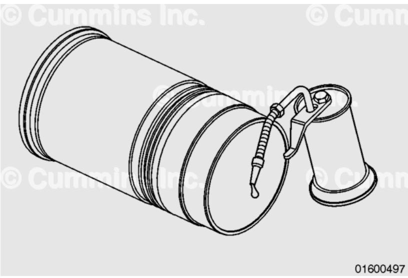
To reduce the possibility of engine damage, the rings must fit correctly in the piston.

Use piston ring compressor, Part Number 4918898.

Install the ring compressor on the piston.

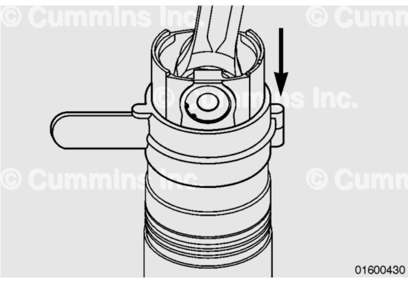
When installing the ring compressor, the counterbore on the tool inner diameter **must** face toward the piston crown.

Lubricate the cylinder liner with clean engine oil.



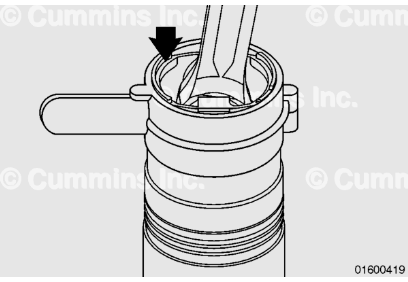
⚠ CAUTION ⚠

Do not allow the connecting rod to impact the liner during the following steps or liner cracking or other damage may occur.



With the cylinder liner in an inverted position, lift the piston and connecting rod so that the counterbore on the ring compressor engages with the underside of the liner.

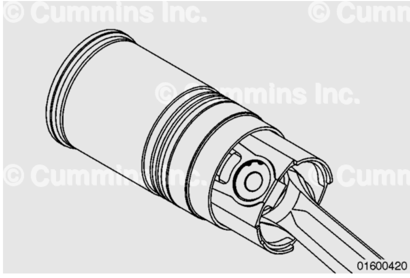
Apply slight downward pressure to insert the piston and connecting rod assembly.



Continue until the piston skirt is approximately level with the upper side of the ring compressor.

Remove the piston ring compressor.

Place the power cylinder assembly on a pallet or bench.

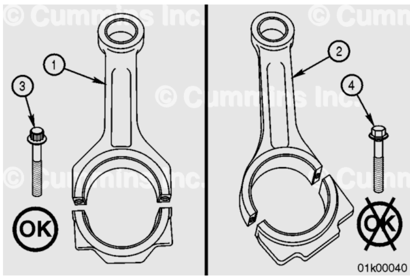


Install

The following steps **must** be used for engines containing the straight split connecting rod (1)

Use the following procedure for engines containing the angle split rod (2). Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

Note : The straight split connecting rod (1) can be identified by 12 point head capscrews (3).



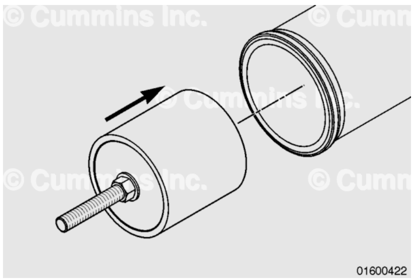
⚠ CAUTION ⚠

Do not allow the connecting rod to impact the liner during the following steps, or liner cracking or other damage may occur.

Remove any excess oil from the liner bore.

Insert the plunger, power cylinder install/removal tool, Part Number 4919227, into the liner.

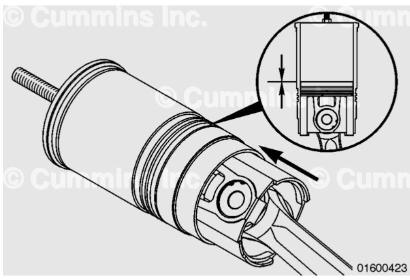
The top face of the tool should be approximately 20 mm [0.8 in] below the top of the liner.



Push the piston and connecting rod into the cylinder liner until the piston contacts the bottom of the plunger.

Tighten the plunger to lock the plunger into the liner.

Torque Value: 125 n•m [92 ft-lb]

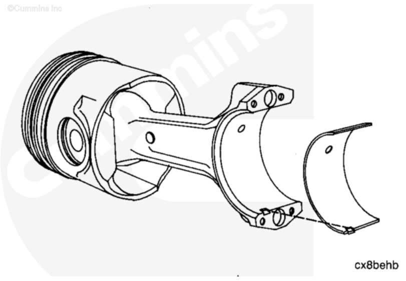


Use a lint-free cloth to clean the connecting rod and bearing shells.

Install the connecting rod bearing.

Position the tang as shown in the illustration. The end of the bearing **must** be even with the cap mounting surface.

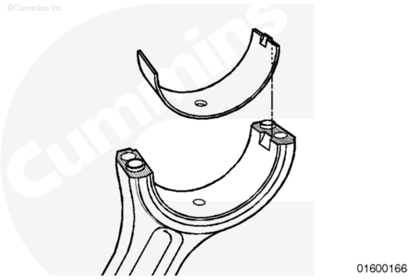
The bearings **must** be installed in their original location if new bearings are **not** used. All of the rod bearings are identical.



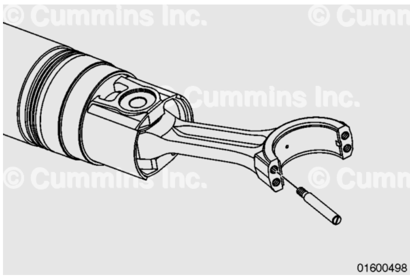
⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the back side of the bearing shells.

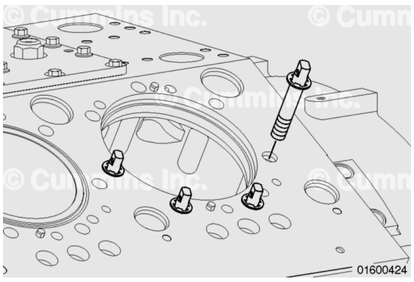
Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086.



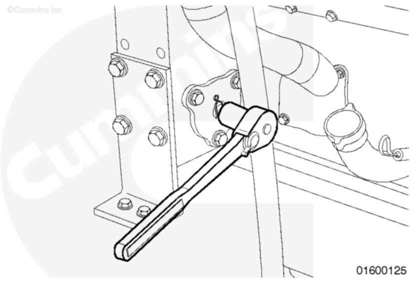
Install the four connecting rod guide studs, Part Number 4918922, into the connecting rod.



Insert the four block guide studs into the block at the locations shown until the circlip contacts the block face.

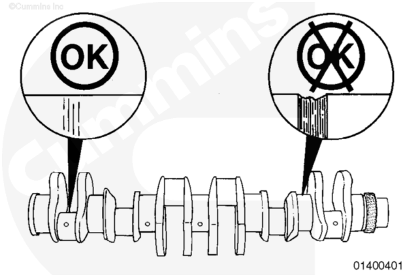


Rotate the crankshaft until the journal for the connecting rod being installed is at top dead center and centered in the cylinder bore.



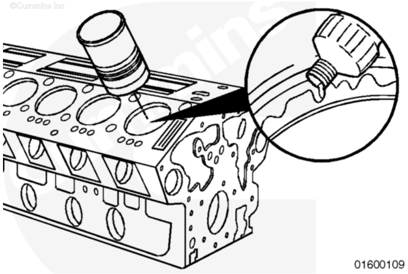
Note : The crankshaft is shown out of the engine for clarity.

Check the crankshaft rod journals for damage.



Do **not** use petroleum-based oil, as this will cause swelling of the packing ring.

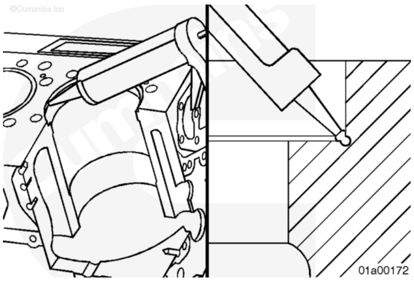
Use vegetable oil to lubricate the inside diameter of the packing ring bores.



Apply a 2 mm [0.079 in] bead of gasket sealant, Part Number 4918771, to the corner of the cylinder block counterbore.

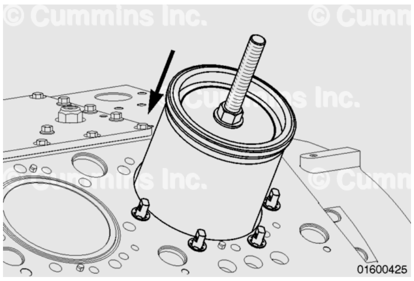
The end of the bead **must** overlap the beginning of the bead by approximately 6 mm [0.236 in].

Install the cylinder liner within 15 minutes of applying the sealant.



⚠ WARNING ⚠

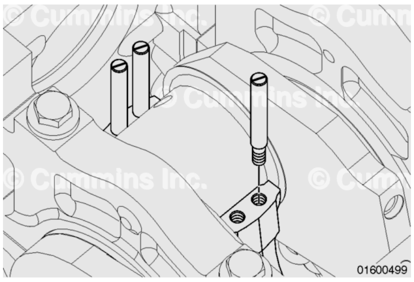
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

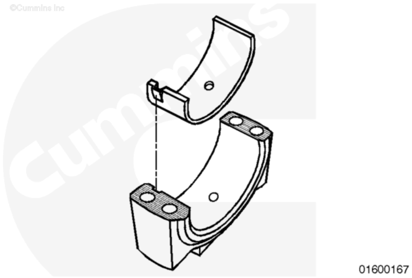


The connecting rod **must not** contact the block and **must** be correctly aligned with the crankshaft journal during installation.

With assistance, or with a hoist, slowly lower the power cylinder into the cylinder bore until the connecting rod contacts the crankshaft journal.

Remove the connecting rod guide studs.



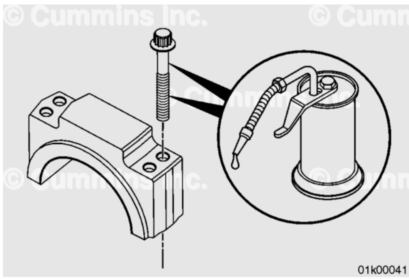


⚠ CAUTION ⚠

The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if they are mixed.

Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

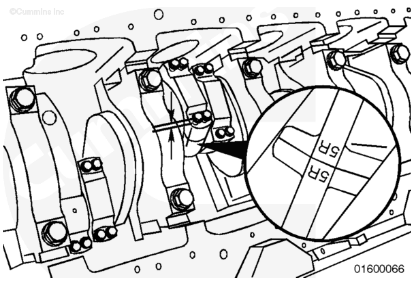
Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086



Lubricate the capscrew head flange with clean gear oil (SAE W140).

Lubricate the capscrew threads with clean engine oil (SAE 15W-40).

Install the capscrews into the caps.



⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.

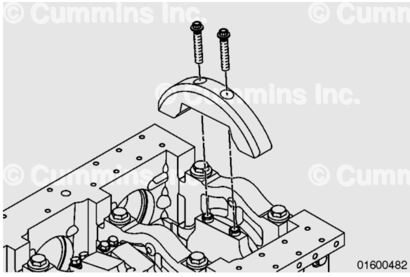
⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backwards, engine damage will occur.

Install the connecting rod cap. Tighten the capscrews alternately and evenly to pull the cap over the dowel pins.

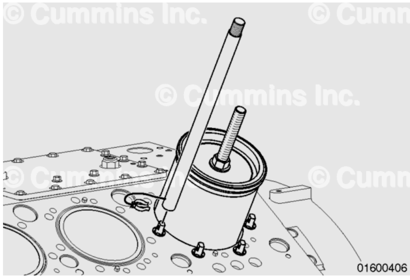
Do **not** tighten the capscrews at this stage.

If a counterweight was previously removed, follow the instructions in the Clean and Inspect for Reuse section. The counterweight should be installed at this time. Refer to Procedure 001-016 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html)



Attach the four frame tube legs to the lower guide studs and secure them with locking pins.

The fixture used in the following steps is used to aid with the alignment of the liner into the block.

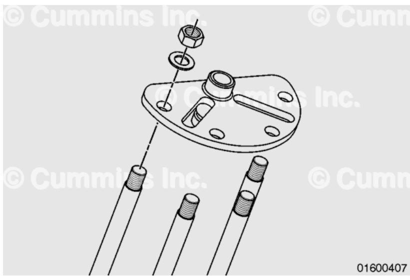


⚠ CAUTION ⚠

Do not overtighten the nuts or tool damage may occur.

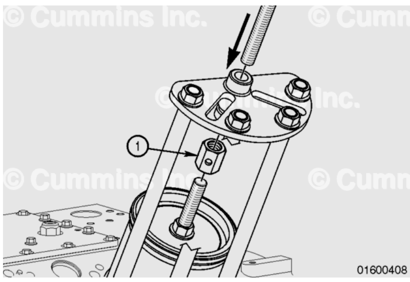
Attach the top plate with the boss facing upward.

Secure the plate with the four nuts and washers provided.



During the following tool assembly, and until instructed otherwise, the upper nut and thrust washer **must not** be fitted.

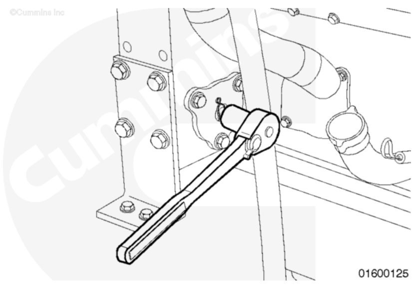
Insert the threaded bar through the top plate and attach it to the plunger using the hexagonal coupling (1).



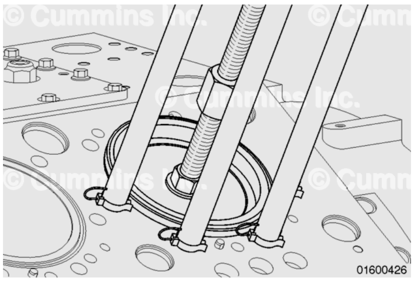
During the following procedure, the cylinder liner should be drawn into the block. If no movement is observed, stop and investigate what is preventing the liner from being drawn into the block.

Do **not** attempt to draw the liner flush with the block.

Use an engine barring mechanism to rotate the engine to lower the power cylinder assembly into the block.



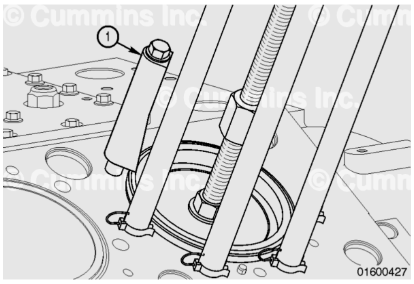
Lower the power cylinder until the liner contacts the press fit area of the block, as illustrated.



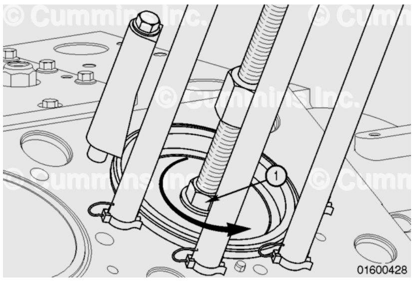
Do **not** attempt to press the liner into the press fit area using liner clamps.

To prevent liner movement during plunger extraction, use a liner clamp set (1), Part Number 3822503, and two cylinder head capscrews to hold the liner in position, as illustrated.

Hand-tighten the capscrews.

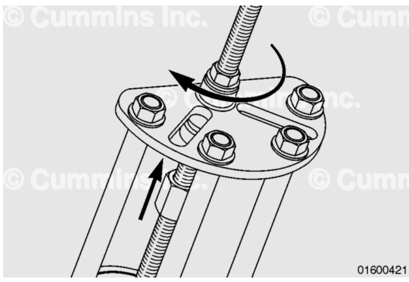


Loosen the plunger nut (1).



Fit the thrust washer and nut to the threaded rod.

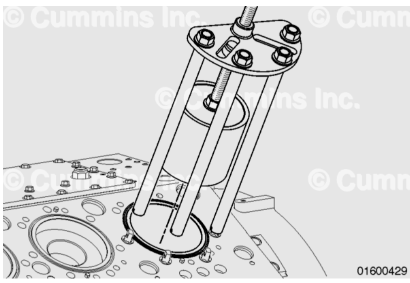
Continue rotating the nut **clockwise** to draw the plunger from the cylinder liner to a position where it can be easily removed by hand.



Release the locking pins from the block guide studs and remove the power cylinder tool as a complete assembly.

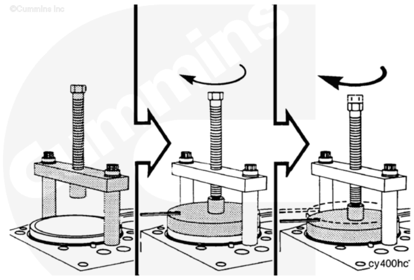
Remove the block guide studs.

Remove the liner clamps.

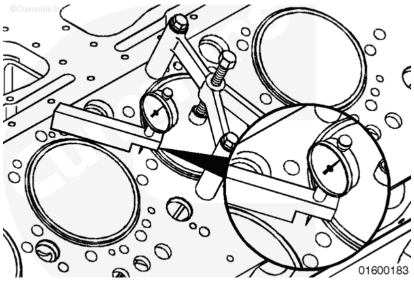


Use a liner installation tool, Part Number 3163329, or equivalent, to fully install the cylinder liner. Refer to Procedure 001-028 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)



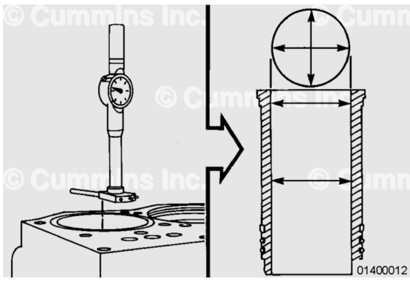
Measure the cylinder liner protrusion.



Cylinder Liner Protrusion		
mm		in
0.15	MIN	0.006
0.22	MAX	0.009

If the protrusion is **not** correct, the liner **must** be removed. The protrusion can be adjusted using seal rings and/or machining the counterbore ledge. Contact a Cummins® Authorized Repair Location if machining is required.

Use a dial bore gauge to measure the inside diameter of the cylinder liner at the top and middle of the cylinder liner.

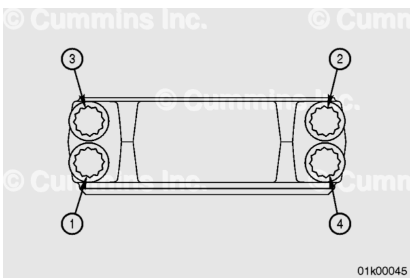


Perform two measurements at each location. The measurements **must** be 90-degrees apart.

New Cylinder Liner Inner Diameter		
mm		in
158.75	MIN	6.250
158.78	MAX	6.251

The inside diameter **must not** be more than 0.076 mm [0.003 in] out of round.

Note : Use the following steps for engines containing the straight split connecting rod only. Use the following procedure for engines containing the angle split connecting rod. Refer to Procedure 001-054 in Section 1 (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html). The straight split connecting rod can be identified by 12 point head capscrews.



Tighten the four connecting rod bearing cap capscrews.

Use sequence 1,2,3,4 on each connecting rod when tightening the connecting rod capscrews. See the illustration.

Torque Value:

1. 115 n•m [85 ft-lb]
2. Repeat step 1
3. Loosen all capscrews at least two revolutions
4. 95 n•m [70 ft-lb]
5. Repeat step 4
6. Advance 60 degrees

Use the following tightening procedure if accessing the connecting rod cap through the hand hole cover and a torque angle dial is inaccessible.

Prior to applying this process, be sure that all capscrews have been tightened by hand as far as possible to prevent thread damage or mis-engagement.

Use the following steps and sequence to tighten the capscrews.

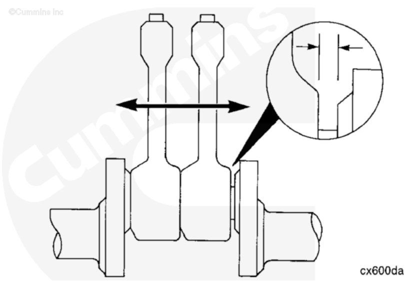
Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the illustration.

Torque Value:

1. 115 n•m [85 ft-lb]
2. Repeat Step 1
3. Loosen all capscrews at least two revolutions
4. 95 n•m [70 ft-lb]
5. Repeat Step 4
6. 205 n•m [151 ft-lb]
7. Repeat Step 6

Note : The torque only method must only be used when a torque angle dial is inaccessible.

Check the side clearance between the rod and the crankshaft. Move the rods as far as possible toward the crankshaft web.



Measure the side clearance with a feeler gauge.

Connecting Rod Side Clearance		
mm		in
0.30	MIN	0.012
0.51	MAX	0.020

Both rods **must** move freely from side-to-side.

Finishing Steps

⚠ **WARNING** ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-046-tr.html)
- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Install the lubricating oil pan suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Install the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Install the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)
- Install the new oil filter elements. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Fill the engine with clean engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Perform the engine run-in procedure. Refer to Procedure 014-002 in Section 14. (/qs3/pubsys2/xml/en/procedures/56/56-014-002.html)